

GIRRAWEE HIGH SCHOOL

YEAR 12 – Task 3 - 2004

MATHEMATICS (Extension)

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a *new* sheet of paper.

Question 1 (21 marks)

Marks

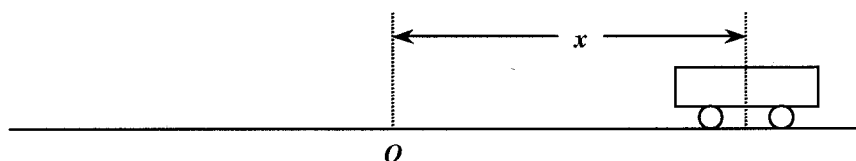
- | | | |
|---|---|---|
| (a) Differentiate with respect to x | (b) Find | |
| (i) $\sin^{-1}(2x)$ | (i) $\int_0^1 \frac{dx}{1+x^2}$ | 3 |
| (ii) $\cos^{-1}(x^2)$ | (ii) $\int_1^2 \frac{dx}{\sqrt{4-x^2}}$ | 3 |
| (iii) $\tan^{-1}\left(\frac{x}{2}\right)$ | (iii) $\int \frac{dx}{1+7x^2}$ | 3 |
| | (iv) $\int \frac{1}{\sqrt{9-4x^2}} dx$ | 3 |

Question 2 (13 marks)

Marks

- (a) A trolley is moving about the origin O . The displacement, x metres, of the trolley from the O at time t seconds is given by,

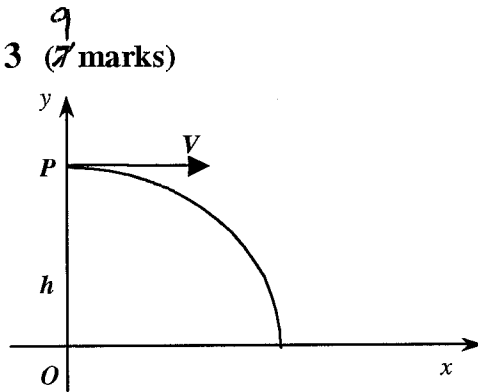
$$x = 6 \sin\left(2t + \frac{\pi}{4}\right).$$



- | | |
|---|-----|
| (i) Show that it is simple harmonic motion. | 2 3 |
| (ii) State the period and amplitude of the motion. | 2 |
| (iii) Find the velocity of the trolley when $t = 0$. | 2 |
| (iv) Find the first time after $t = 0$ when the center of the trolley is at $x = 3$. | 2 3 |
| (b) A particle is moving according to $\ddot{x} = 2x$. Initially it is stationary at $x = 1$. Find v^2 as a function of x . | 3 |

Question 3 (7 marks)

Marks



- (a) A particle is projected horizontally from a point P, h metres above O, with A velocity of V metres per second. The equation of motion of the particle are,

$$\ddot{x} = 0 \quad \& \quad \ddot{y} = -g$$

Using calculus, show that the position of the particle at time t is

given by. $x = Vt$, $y = h - \frac{1}{2}gt^2$

24

- (b) A canister containing a life raft is dropped from a plane to a stranded sailor. The plane is traveling at a constant velocity of 216 km/h, at a height of 120 metres above sea level, along a path that passes above the sailor.

- (i) Convert 216 km/h to m/s.

2

- (ii) How long will the canister take to hit the water?
(Take $g = 10 \text{ m/s}^2$.)

3

Question 4 (10 marks)

- (a) A particle is moving in SHM along the x -axis and its velocity v m/s at the position x metres is given by $v^2 = 16 + 4x - 2x^2$.

- (i) Show that the acceleration at any time t is given by, $\ddot{x} = -2(x - 1)$.

2

- (ii) Find the center and period of the motion.

2

- (iii) Find the extreme points of the motion and hence find the amplitude of the motion.

2

- (b) A spherical balloon is being inflated so that its rate of increase of volume $\frac{dV}{dt}$ is $5 \text{ cm}^3/\text{s}$. Find the rate of increase $\frac{dS}{dt}$ of its surface area S when the radius of the balloon is 8 cm.

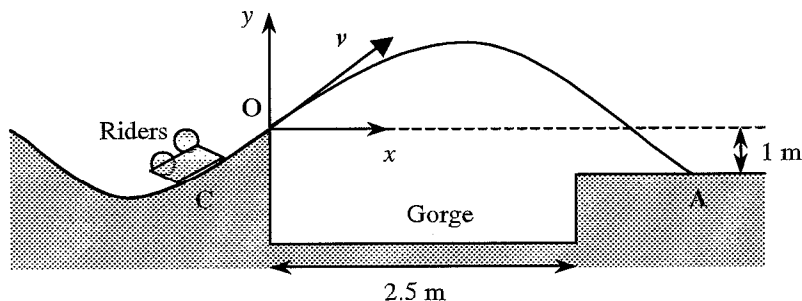
4

19

Question 5 (13 marks)

Marks

- (a) Two riders are in a snow-mobile C, which is about to make a jump across a gorge of width 2.5m. At the edge of the gorge, at the point O, the speed v of the car is 5 m/s and the angle of projection $\theta = 30^\circ$ above the horizontal.



Taking the co-ordinate axes at O and $g = 10 \text{ ms}^{-2}$,

- (i) Show that the car C will be able to make the jump across the gorge and find the horizontal distance of the point A of landing from the point of projection O. 36
- (ii) Find the angle of the velocity with the horizontal at the point A. 34
- (b) In a colony of 2000 birds, the number N infected with a disease at time t is given by $N = \frac{2000}{1 + ke^{-2000t}}$, where k is a constant and t is in years.
- (i) Show that eventually all the birds will be infected. 2
- (ii) If at $t = 0$, one bird was infected, after how many days will 1000 birds be infected? Give your answer to the nearest hour (assume 365.25 days in a year). 34
- (iii) Show that $\frac{dN}{dt} = N(2000 - N)$. 23

Nov 12 - TASK 3 (2004)

* Mathematics (Extension)

Q1 a) (i) $\frac{1}{\sqrt{1-4x^2}} \cdot 2$

$= \frac{2}{\sqrt{1-4x^2}}$

(ii) $= -\frac{1}{\sqrt{1-x^2}} \cdot 2x$

$= \frac{-2x}{\sqrt{1-x^2}}$

(iii) $= \frac{2}{2^2 + x^2}$

$= \frac{2}{4 + x^2}$

b) (i) $= \left[\tan^{-1} x \right]_0$

$= \frac{\pi}{4} - 0$

$= \frac{\pi}{4}$

(ii) $= \left[\sin^{-1} \frac{x}{2} \right]_1$

$= \sin^{-1} \frac{2}{2} - \sin^{-1} \frac{1}{2}$

$= \frac{\pi}{2} - \frac{\pi}{6}$

$= \frac{\pi}{3}$

(iii) $= \int \frac{dx}{1+(\sqrt{7}x)^2}$

$= \frac{1}{\sqrt{7}} \tan^{-1}(\sqrt{7}x) + C$

(iv) $= \int \frac{dx}{\sqrt{3^2 - 2x^2}}$

$= \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C$

Q2 a) (i) $x = 6 \sin\left(2t + \frac{\pi}{4}\right)$

$\dot{x} = 12 \cos\left(2t + \frac{\pi}{4}\right)$

$\ddot{x} = -24 \sin\left(2t + \frac{\pi}{4}\right)$

$\ddot{x} = -4 \times 6 \sin\left(2t + \frac{\pi}{4}\right)$

$\ddot{x} = -2^2 x$

which is in the form $\ddot{x} = -n^2 x$ for a particle to move in SHM.

(ii) $T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$ seconds

amplitude = 6 m

(iii) when $t=0$

$\dot{x} = 12 \cos\left(2(0) + \frac{\pi}{4}\right)$

$= 12 \left(\frac{1}{\sqrt{2}}\right)$

$= \frac{12}{\sqrt{2}} = 6\sqrt{2} \text{ m/s}$

(iv) when $x=3$

$3 = 6 \sin\left(2t + \frac{\pi}{4}\right)$

$\frac{1}{2} = \sin\left(2t + \frac{\pi}{4}\right)$

$2t + \frac{\pi}{4} = \frac{\pi}{6}, \frac{5\pi}{6}, \dots$

$2t = -\frac{\pi}{12}, \frac{7\pi}{12}, \dots$

$t = -\frac{\pi}{24}, \frac{7\pi}{24}, \dots$

$t = \frac{7\pi}{24}$ is first time particle is at $x=3$

b) $\ddot{x} = 2x \rightarrow a = 2x$

$v^2 = 2 \int a dx$

$v^2 = 2 \int 2x dx$

$v^2 = 2x^2 + C$

when $x=1, v=0$

$0 = 2(1)^2 + C$

$C = -2$

$v^2 = 2x^2 - 2$ (or) $v^2 = 2(x^2 - 1)$

Q3

$$\ddot{x} = 0$$

a)

$$\dot{x} = C_1$$

$$\text{when } t=0, \dot{x} = V$$

$$\therefore C_1 = V$$

$$\dot{x} = V$$

$$x = Vt + C_2$$

$$\text{when } t=0, x=0$$

$$\therefore C_2 = 0$$

$$\therefore \underline{x = Vt}$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C_3$$

$$\text{when } t=0, \dot{y}=0$$

$$\therefore C_3 = 0$$

$$\dot{y} = -gt$$

$$y = -\frac{1}{2}gt^2 + C_4$$

$$\text{when } t=0, y=h$$

$$\therefore C_4 = h$$

$$\underline{y = h - \frac{1}{2}gt^2} \quad (4) \quad \text{then}$$

b)

$$\frac{dV}{dt} = 5 \text{ cm}^2/\text{s}$$

$$V = \frac{4}{3}\pi r^3$$

$$\therefore \frac{dV}{dr} = 4\pi r^2$$

$$\therefore \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$5 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{5}{4\pi r^2}$$

$$\frac{ds}{dt} = \frac{ds}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times \frac{5}{4\pi r^2}$$

$$\underline{\frac{ds}{dt} = \frac{10}{r}}$$

$$\text{when } r=8, \frac{ds}{dt} = \frac{10}{8} \quad (4)$$

$$\frac{ds}{dt} = \underline{\underline{1\frac{1}{4} \text{ m/s}}}$$

$$S = 4\pi r^2$$

$$\frac{ds}{dr} = 8\pi r$$

$$\text{b) (i) } 216 \text{ km/h} \\ = \underline{60 \text{ m/s.}} \quad (2)$$

$$\text{(ii) From (i) } y = 120 - \frac{1}{2}(10)t^2$$

$$\text{when } y=0$$

$$0 = 120 - 5t^2$$

$$5t^2 = 120$$

$$t^2 = 24$$

$$t = \pm \sqrt{24} = \pm 2\sqrt{6}$$

$$\text{only } +2\sqrt{6} \text{ secs.} \quad (3)$$

Q4

$$v^2 = 16 + 4x - 2x^2 \quad v^2 = 2(8 + 2x - x^2)$$

$$\text{(i) } a = \frac{1}{2} \frac{d(v^2)}{dx}$$

$$a = \frac{d}{dx} \left(\frac{1}{2} (16 + 4x - 2x^2) \right)$$

$$= \frac{d}{dx} (8 + 2x - x^2)$$

$$a = 2 - 2x$$

$$a = -2(x-1) \quad \text{or } \ddot{x} = -2(x-1) \quad (2)$$

$$\text{(ii) } \therefore n = \sqrt{2}$$

Centre of Motion

$$T = \frac{2\pi}{\sqrt{2}}$$

$$\text{is } \underline{x=1 \text{ m}} \quad (2)$$

$$T = \sqrt{2} \pi \text{ seconds}$$

$$\text{as } \ddot{x} = 0 \text{ when } x=1$$

$$\text{(iii) when } v=0$$

$$0 = 2(8 + 2x - x^2)$$

$$= 2(4-x)(2+x)$$

$$\therefore \text{end points are } x=2, 4$$

Amplitude

$$\text{is } \underline{3 \text{ m}} \quad (2)$$

Q5

$$\ddot{x} = 0$$

$$\dot{x} = \frac{5\sqrt{3}}{2}$$

$$x = \frac{5\sqrt{3}}{2}t$$

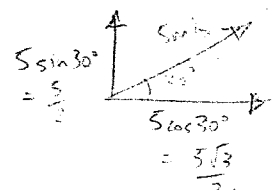
when

$$y = -1 \text{ m}$$

$$-1 = -5t^2 + \frac{5}{2}t$$

$$10t^2 - 5t - 2 = 0$$

INITIAL Vel Diagram



$$\ddot{y} = -10$$

$$\dot{y} = -10t + \frac{5}{2}$$

$$y = -5t^2 + \frac{5}{2}t$$

$$t = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(10)(-2)}}{2(10)}$$

$$= \frac{5 \pm \sqrt{25+80}}{20}$$

$$t = \frac{5 \pm \sqrt{105}}{20}$$

$$t = \frac{5 + \sqrt{105}}{20} \quad \text{only}$$

$$\text{sub } t = \frac{5 + \sqrt{105}}{20} \text{ into } x = \frac{5\sqrt{3}}{2}t$$

$$x = \frac{5\sqrt{3}}{2} \left(\frac{5 + \sqrt{105}}{20} \right)$$

$$= \frac{25\sqrt{3} + 5\sqrt{315}}{40}$$

$$= \frac{5\sqrt{3} + \sqrt{315}}{8}$$

Greater than

$$x = 2.5 \text{ m}$$

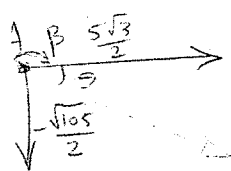
i.e. car will

cross the line

(6)

cont. (i) when $y = -1$; $x = \frac{5 + \sqrt{105}}{20}$ from (i)

$$\dot{x} = \frac{5\sqrt{3}}{2}; \quad \dot{y} = -10\left(\frac{5 + \sqrt{105}}{20}\right) + \frac{5}{2}$$



$$\dot{y} = -\frac{\sqrt{105}}{2}$$

$$\tan \theta = \frac{\sqrt{105}/2}{5\sqrt{3}/2}$$

or $\beta = 180 - 49^\circ 48'$
 $\beta = 130^\circ 12'$
 $\tan \theta = \frac{\sqrt{105}}{5\sqrt{3}}$
 $\theta = 49^\circ 48'$

b) $N = \frac{2000}{1 + ke^{-2000t}}$

(i) when $t \rightarrow \infty$, $e^{-2000t} \rightarrow 0$

$$\therefore N = \frac{2000}{1+0}$$

$$N \rightarrow 2000$$

(ii) when $t = 0$; $N = 1$

$$1 = \frac{2000}{1 + ke^{-2000(0)}}$$

$$1 = \frac{2000}{1+k}$$

$$1+k = 2000$$

$$\therefore k = 1999$$

⊛ $N = \frac{2000}{1 + 1999e^{-2000t}}$

when $N = 1000$

$$1000 = \frac{2000}{1 + 1999e^{-2000t}}$$

$$1 + 1999e^{-2000t} = 2$$

$$1999e^{-2000t} = 1$$

$$e^{-2000t} = \frac{1}{1999}$$

$$\ln e^{-2000t} = \ln \frac{1}{1999}$$

$$-2000t = \ln \frac{1}{1999}$$

$$\therefore t = \frac{\ln(1/1999)}{-2000}$$

$$t = 0.0038 \text{ years}$$

$$t = 1.388 \text{ days}$$

$$\therefore t \approx 33.3 \text{ hrs.}$$

(iii) If $N = 2000(1 + ke^{-2000x})$

$$\frac{dN}{dt} = -2000(1 + ke^{-2000t})^{-2} \cdot -2000ke^{-2000t}$$

$$= \frac{2000^2 ke^{-2000t}}{(1 + ke^{-2000t})^2} \quad (1)$$

but,

$$N + Nke^{-2000t} = 2000$$

$$Nke^{-2000t} = 2000 - N$$

$$ke^{-2000t} = \frac{2000 - N}{N} \quad (2)$$

Sub (2) into (1)

$$\frac{dN}{dt} = \frac{2000^2 \left(\frac{2000 - N}{N} \right)}{\left(1 + \frac{2000 - N}{N} \right)^2}$$

$$= \frac{2000^2 \left(\frac{2000 - N}{N} \right)}{\frac{2000^2}{N^2}}$$

$$\boxed{\frac{dN}{dt} = N(2000 - N)}$$

(3)

Q5) (all mode)

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$x = \frac{5\sqrt{3}}{2}$$

$$\dot{y} = -10t + \frac{5}{2}$$

$$x = \frac{5\sqrt{3}}{2}t$$

①

$$y = -5t^2 + \frac{5}{2}t$$

②

$$t = \frac{2x}{5\sqrt{3}}$$

③

sub ③ into ②

$$y = -\left(\frac{2x}{5\sqrt{3}}\right)^2 + \frac{5}{2}\left(\frac{2x}{5\sqrt{3}}\right)$$

Cartesian
Eqn.

$$y = -\frac{4}{15}x^2 + \frac{2}{\sqrt{3}}x$$